

The UAb WISEflow Automation & Integrity Suite

A Case Study in Engineering Scalable and
Secure Digital Assessment

The Challenge: Delivering Digital Assessment with Excellence at Scale

Universidade Aberta, as a leader in distance learning, faces a unique operational challenge: managing thousands of digital assessments annually. This requires a system that is not only efficient and scalable but also guarantees the highest standards of academic integrity and inclusivity for all students.



Operational Scale:
Managing the entire lifecycle of hundreds of concurrent assessments ('flows').



Data Integrity: Ensuring a single source of truth for user roles, participant lists, and assessment results across multiple systems.

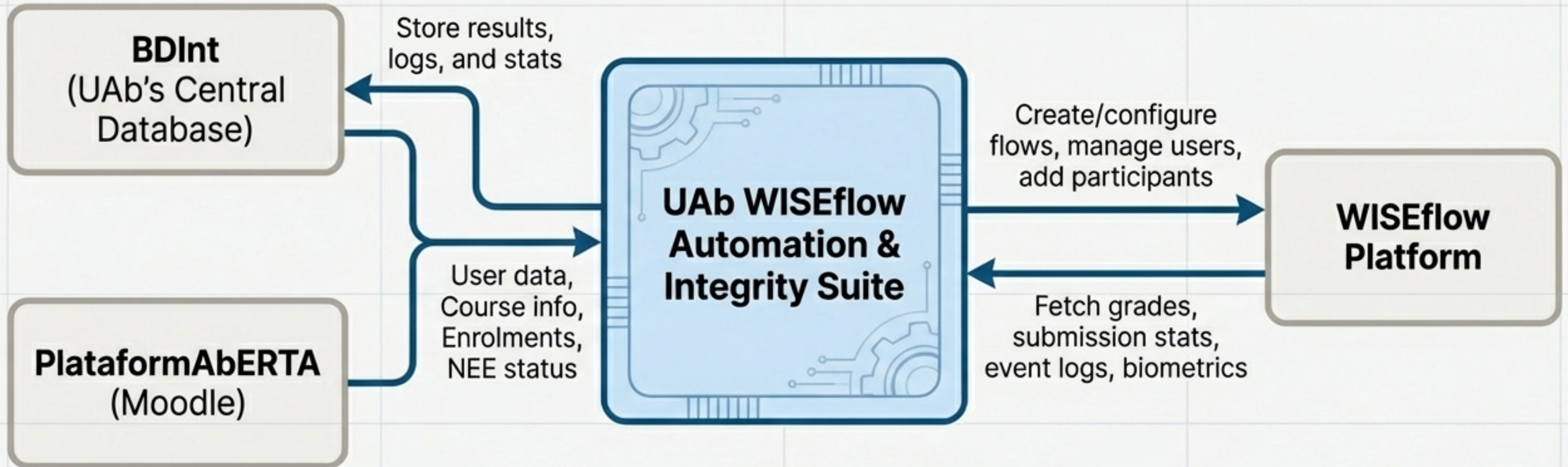


Academic Integrity:
Proactively monitoring assessment sessions to ensure fairness and prevent misconduct.



Inclusivity: Systematically accommodating students with special educational needs (NEEs) without manual intervention.

The Solution: An Integrated Automation Engine



The **Suite** is the **central nervous system**, orchestrating the entire assessment lifecycle by interfacing directly with the **WISEflow API** and **UAb's internal systems**.

Deconstructing the Engine: Four Pillars of Automation



Flow Lifecycle Management

End-to-end automation from creation and configuration to activation and archiving.



User & Participant Integrity

Synchronised and rule-based management of all user accounts, roles, and assessment enrolments.



Automated Reporting & Data Sync

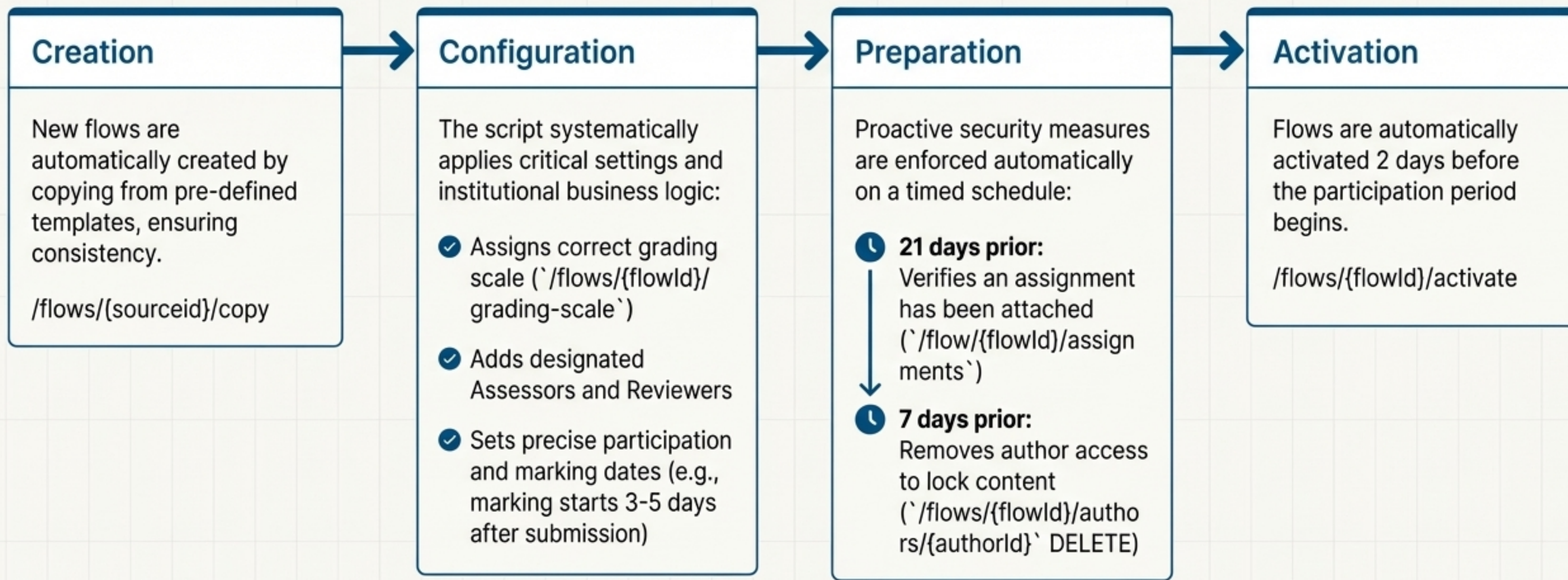
Proactive notifications and seamless extraction of grades, statistics, and biometric markers.



Proactive Integrity Monitoring

Real-time, intelligent analysis of student behaviour during assessments.

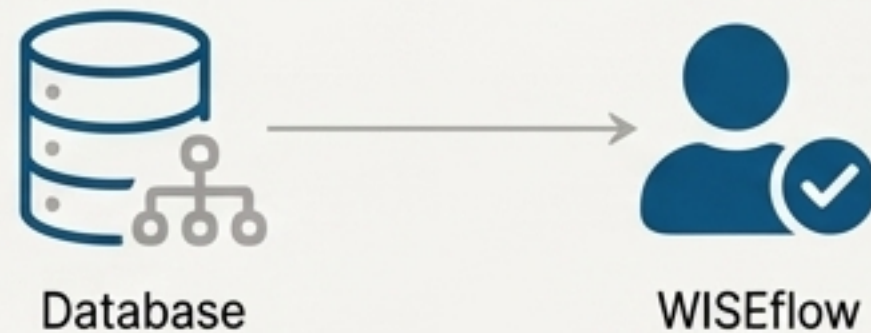
Pillar 1: Automating the Complete Assessment Lifecycle



Pillar 2: Ensuring Every User and Participant is in the Right Place

User Account Synchronisation (`sync\_users`)

Core Concept: Single Source of Truth: User accounts in WISEflow are synchronised with UAb's central database.



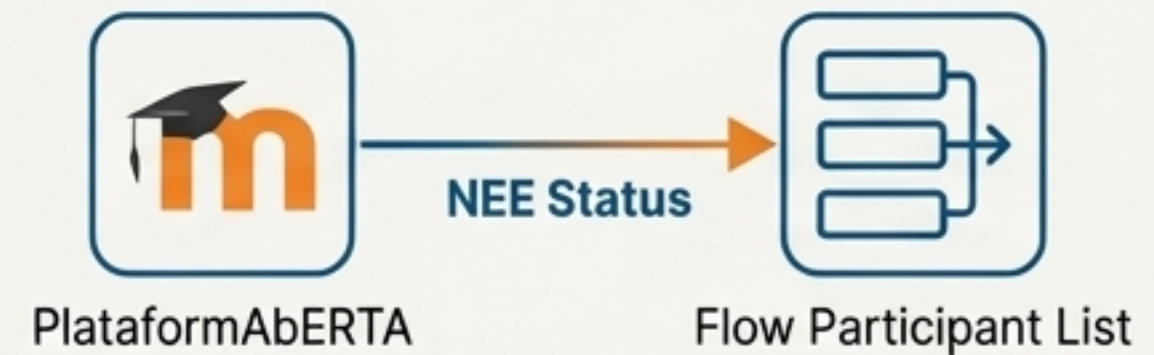
Lifecycle Management

- New students are created (/license/user POST).
- Inactive students are deactivated (/users/{userId} PUT with loginDeactivated: true).
- Staff roles are updated (/users/{userId}/roles POST), granting appropriate permissions for Teachers vs. Tutors.

Why it matters: Eliminates manual account creation, prevents permission errors, and ensures data accuracy.

Participant Enrolment & Accommodations (`sync\_parts`)

Core Concept: Automated Enrolment: Adds and removes participants from flows based on course enrolment data.



Inclusive by Design

- The script pulls Special Educational Needs (NEE) status from PlataformAbERTA (Moodle).
- **Automated Time Extensions:** For students with NEE status, it automatically extends submission deadlines (/flows/{flowId}/participants/{partId}/dates PATCH) by their allotted extra time (e.g., `xtrT * 60` seconds).

Why it matters: Guarantees equitable assessment conditions and removes the administrative burden of managing individual accommodations.

Pillar 2: Maintaining Data Fidelity Through Proactive Rectification

Focus Case: `rectify\_assessor\_type`

Problem

Assessor profiles on flows, created from templates, could become outdated or misaligned with current staff assignments.



Automated Solution

The script runs periodically on all future-dated flows:



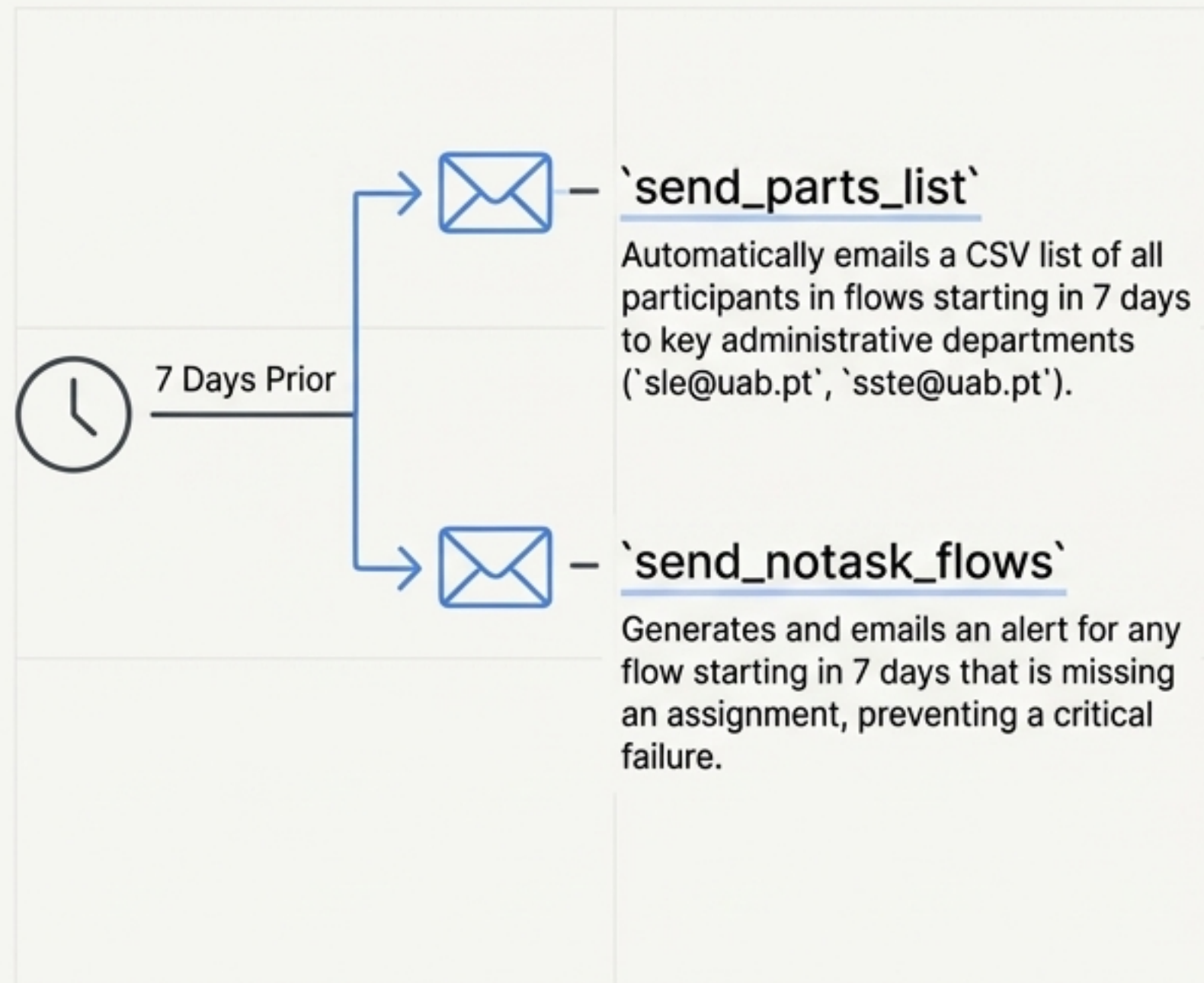
Why it matters

This ensures absolute accuracy in assessor assignments, which is critical for correct marking workflows, auditing, and preventing administrative problems. It demonstrates a commitment to self-healing data.

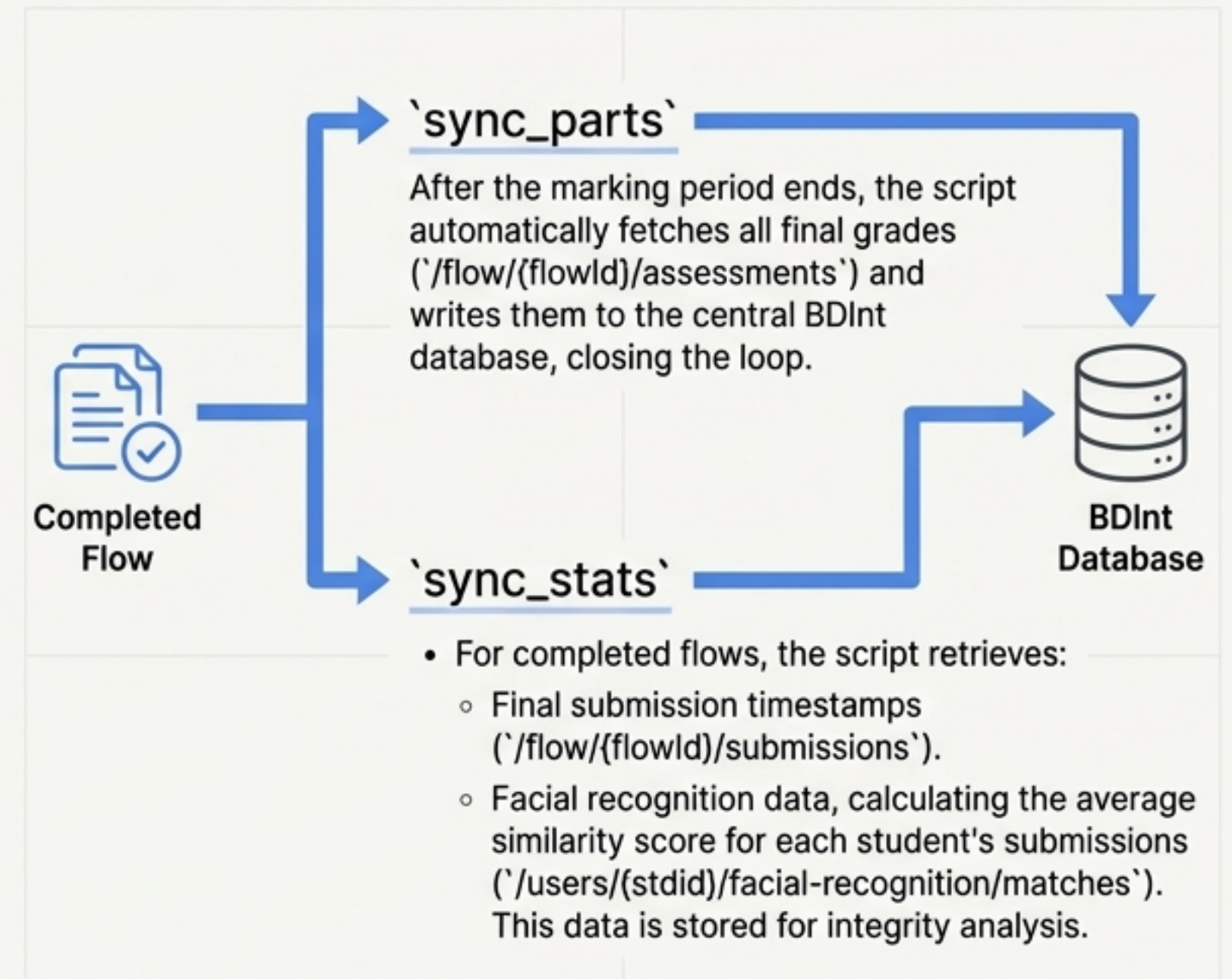


Pillar 3: Delivering Timely Insights Through Automated Reporting & Data Sync

Proactive Notifications



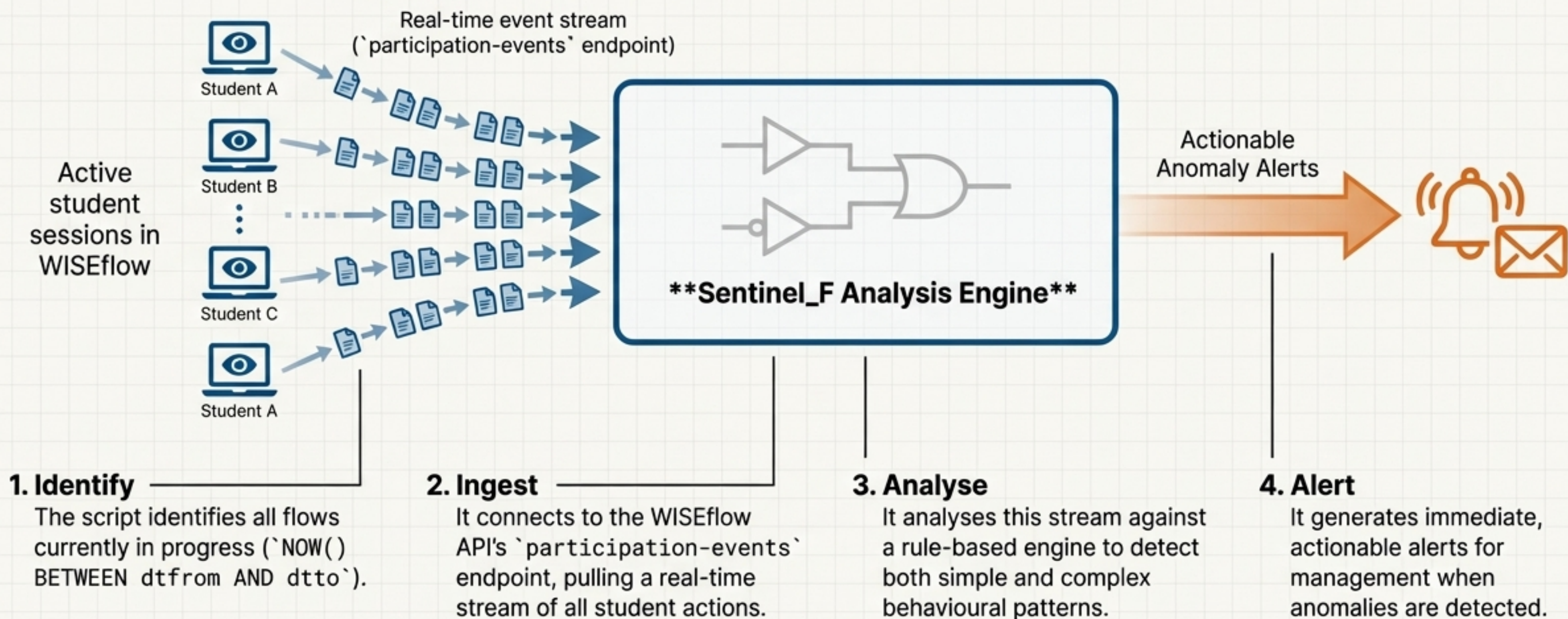
Post-Assessment Data Harvesting





Pillar 4: Sentinel_F, a Vigilant System for Real-Time Academic Integrity Monitoring

Core Purpose Statement: Sentinel_F moves beyond simple automation to provide active, intelligent oversight of student behaviour during live digital assessments.



Tapping the Event Stream: The Data Behind Sentinel_F's Analysis

Data Source: The script continuously polls the /participation-events' API endpoint for all active flows, ingesting thousands of events and storing them in a dedicated database for analysis.



FLOW_OPENED

When a student begins their session.



DEVICE_MONITOR_SCREENSHOT

Logs screen captures from the proctoring system.



ASSIGNMENT_ACCESSED

Accessing exam documents.



FACIAL_RECOGNITION

Logs facial recognition match attempts.



CHARACTERS_TYPED

Captures character input activity for behavioural analysis.



VOICE_DETECTION

Logs audio events captured during invigilation.



PAPER_SAVED / PAPER_HANDED_IN

Tracks submission progress and completion.



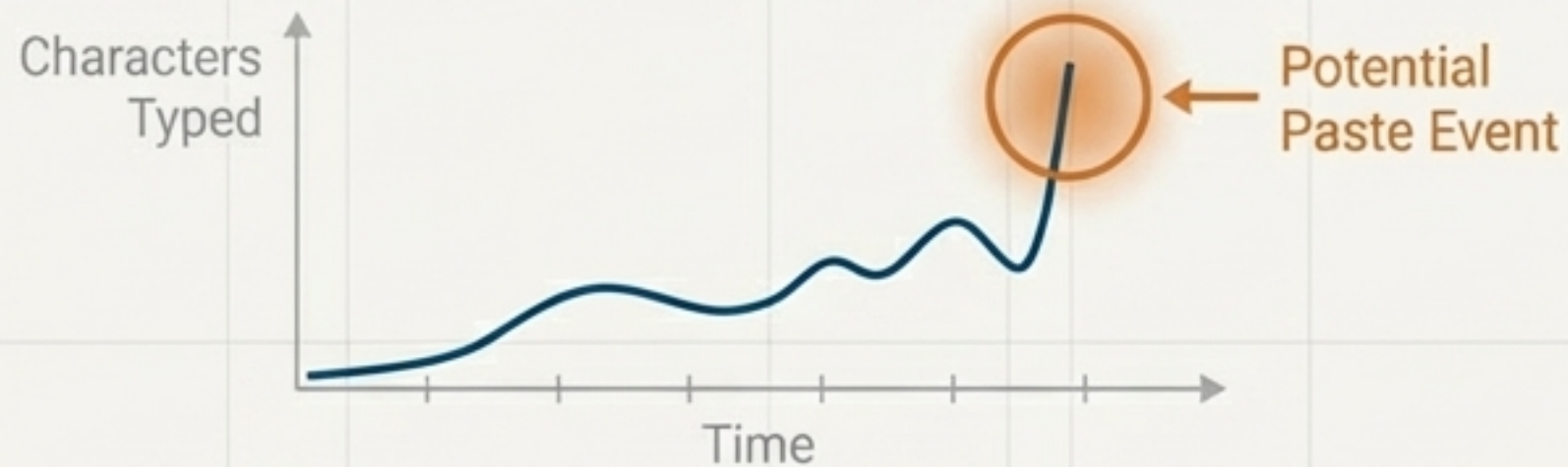
Self-Adapting Feature

Sentinel_F is designed to identify, log, and immediately alert administrators to *any new, uncatalogued event types** returned by the API, ensuring the system evolves with the WISEflow platform.

From Data to Insight: Detecting Complex Behavioural Patterns

Sentinel_F goes beyond single events to analyse relationships between them, identifying 'composite events' that indicate potential academic misconduct.

Sudden Spikes in Typing Speed



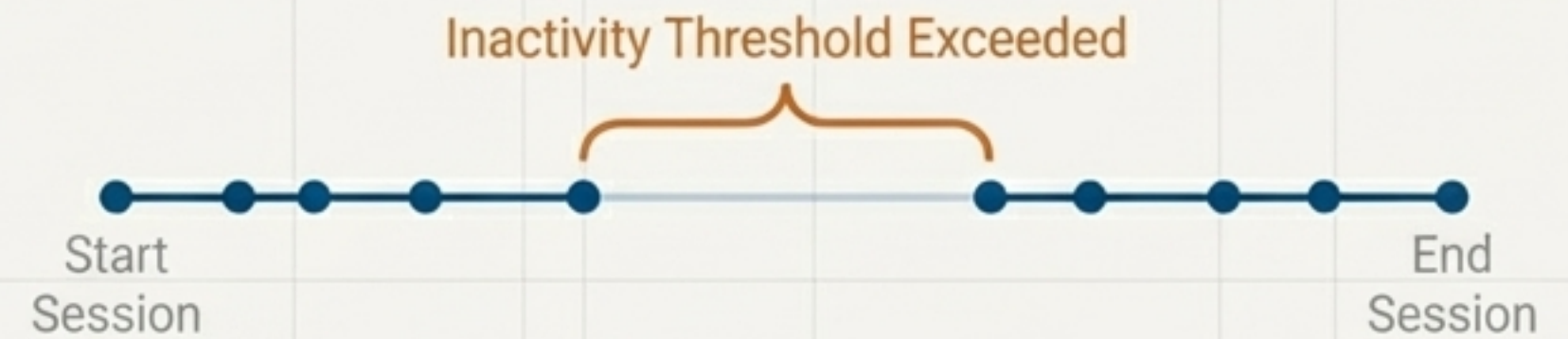
Method

For each student, the script analyses consecutive `CHARACTERS_TYPED` events. It calculates the difference in characters and the difference in time, deriving a `chars_per_second` rate.

Trigger

An **alert** is generated if this rate exceeds a pre-defined threshold, a strong indicator of large-scale text pasting.

Prolonged Inactivity



Method

The script measures the time (`TIMESTAMPDIFF`) between every consecutive event from a single student.

Trigger

If the time gap surpasses a configured threshold (e.g., several minutes), the system generates a synthetic **INACTIVITY alert**, flagging potential disengagement or absence.

Closing the Loop: Actionable Alerts and Management Reporting



Immediate Management Alerts

When a relevant elementary or composite event is detected, a detailed email alert is sent in real-time to a designated management group (``manageT0``, ``manageCC``). The email includes a table with Flow, Student ID, Timestamp, Event Type, and Payload.

Subject: ALERTA: eventos relevantes detectados

Flow	Student ID	Timestamp	Event Type	Payload
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____



Daily Summary Reports

In ``report`` mode, the script generates a daily digest for stakeholders (``reportT0``). This report summarises:

- A count of all relevant events detected that day, by type.
- A list of any flows that concluded with zero student participation.

Event Counts (by Type)



Zero Participation Flows

- Flow ID: [Flow_123]

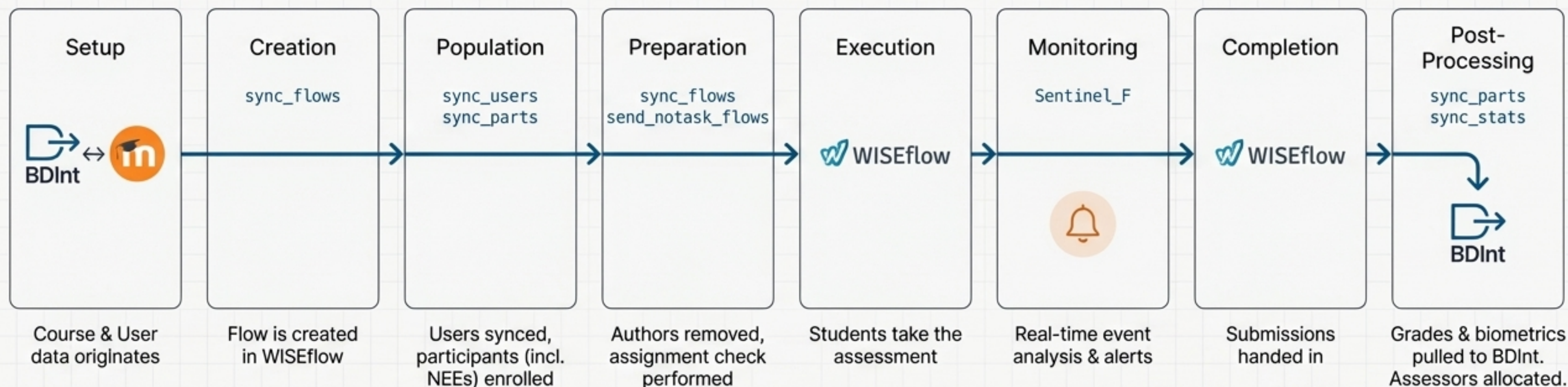


Automated Housekeeping

The system automatically purges raw event data older than 365 days to maintain performance and comply with data retention policies.



The UAb Suite: An Orchestrated Digital Assessment Journey



Each stage is automated by a specific component of the suite, creating a seamless, robust, and reliable process.

A Transformative Impact on Digital Assessment Operations



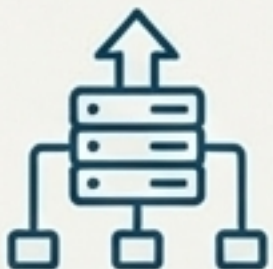
Unprecedented Efficiency

Automation of user management, flow creation, and grade migration has drastically reduced manual administrative hours, freeing up staff for higher-value tasks.



Enhanced Academic Integrity

The move from passive proctoring to proactive, data-driven behavioural analysis with Sentinel_F provides a robust and evidence-based approach to upholding standards.



Robust Scalability & Reliability

The suite provides a resilient framework capable of managing thousands of assessments and users, with proactive checks to prevent common configuration errors.



Systemised Inclusivity

Automated handling of special educational needs ensures that accommodations are applied consistently and accurately, embedding equity into the assessment process itself.

More Than Automation: Engineering Trust in Digital Assessment

The UAb WISEflow Automation & Integrity Suite is a strategic asset. It demonstrates how bespoke integration, built upon a powerful API, can transform a digital assessment platform into a system that is not only efficient but also inherently trustworthy, secure, and equitable.

**This is the blueprint for managing the future of
digital assessment in higher education.**